sensors -- Article

Simulation-Grounded Co-Design with Hardware-Aware Validation of an MFC-Powered AIoT Sensor Node for Autonomous Environmental Monitoring

Kursat Kahya 1,*

1 BILSEM Aerospace and UAV Programme, Türkiye

* Correspondence: kahyakursat1@gmail.com

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Abstract:

This paper presents a simulation-grounded, hardware-aware co-design framework for a battery-free environmental sensor node powered by soil-based microbial fuel cells (MFCs). Unlike purely simulation-driven approaches, the framework integrates literature-calibrated electrochemical modelling with hardware-imperfection-aware validation and an emulated power-path verification stage. The architecture combines MFC energy harvesting (0.3-0.8 V OCV), a 1 F supercapacitor, an ESP32 in 10-minute deep-sleep duty cycles, and a LoRa Ra-02 433 MHz link. An adaptive duty-cycle decision model reduces energy consumption by 47.5% versus always-transmit operation (0.413 vs. 0.787 mJ/cycle). An Isolation Forest anomaly detector achieves F1 = 0.955 +/- 0.022 (5-fold CV, n = 50/class) without labelled training data; subtle-anomaly virtual testing confirms F1 = 0.880 and ROC-AUC = 0.929 under realistic noise. Power-trace emulation with a literature-calibrated MFC voltage profile validates the energy model with MAPE = 8.5% vs. the flat-nominal simulation baseline. A 200-trial Monte Carlo analysis confirms 75.0% energy-positive operation under combined hardware imperfections. The open-source framework provides a reproducible pre-prototyping methodology applicable to any energy-harvesting IoT design.

Keywords: microbial fuel cell; energy harvesting; IoT; anomaly detection; LoRa; Isolation Forest; TinyML; environmental monitoring; ESP32; precision agriculture

1. Introduction

Continuous environmental monitoring in remote agricultural and ecological sites faces a fundamental energy supply challenge. Conventional battery-powered sensor nodes require periodic maintenance, generate electronic waste, and become impractical in locations with limited human access. Solar and wind harvesting are intermittent and site-dependent. Microbial fuel cells (MFCs) offer an alternative: they harvest electrical energy continuously from the electrochemical activity of naturally occurring soil microorganisms, converting the chemical energy stored in organic matter directly into electricity with no moving parts [1,3].

Recent advances have demonstrated that sediment MFCs can sustain wireless sensor nodes capable of measuring soil moisture, temperature, and pH [6,11]. However, the power output of MFCs is inherently variable--governed by substrate concentration, temperature, biofilm maturity, and electrode geometry--making naive always-on operation infeasible [1]. Intelligent duty-cycle management is therefore essential to bridge the gap between intermittent energy availability and reliable sensing.

The emergence of Artificial Intelligence of Things (AIoT) paradigms -- combining edge machine learning with low-power IoT hardware -- opens new possibilities for adaptive sensor nodes [14,15]. Isolation Forest [13], an ensemble-based unsupervised anomaly detection algorithm, is particularly attractive for embedded deployment because it operates without labeled data and has been shown to consume under 160 KB of RAM in optimized implementations on ESP32-class microcontrollers [19,20]. Simultaneously, LoRa (Long Range) modulation enables kilometre-scale wireless communication at sub-milliwatt average power in a single-hop star topology, making it the protocol of choice for battery-free agricultural sensor networks [22,24].

All MFC electrical parameters used in this work are calibrated with values reported in peer-reviewed experimental literature [1,3,6,11]. Training and validation datasets are generated from independent parameter configurations (Scenario A and B) to avoid circular validation.

The main contributions of this study are:

1. A microbial fuel cell-powered, energy-autonomous IoT sensor architecture integrating supercapacitor buffering, ESP32 deep-sleep firmware, and LoRa long-range communication into a single battery-free platform.

2. An embedded adaptive energy management strategy (compact RF decision model as a deployable policy approximating the optimal threshold controller) that dynamically selects sleep, measure, or transmit actions, achieving a 47.5% reduction in average energy consumption vs. a fixed-interval baseline.

3. An unsupervised anomaly detection pipeline (Isolation Forest) requiring no labelled training data, achieving F1 = 0.955 +/- 0.022 (5-fold CV, n = 50/class) on four environmental anomaly types at <160 KB memory footprint -- compatible with ESP32 deployment.

4. A power-trace emulation methodology -- to our knowledge the first to introduce a simulation-to-emulation validation pipeline for MFC-powered AIoT sensor nodes -- that validates the energy model within 8.5% against a programmable power supply MFC profile, bridging the gap between flat-nominal simulation and hardware behaviour (Figure 9).

5. An open-source, reproducible simulation framework for co-design of energy-aware IoT systems prior to physical prototyping, with fully documented parameterisation and a structured Simulation-Emulation-Prototype path toward hardware validation.

2. Related Work

2.1. MFC Energy Harvesting for Wireless Sensors

Logan et al. [1] established the foundational methodology for MFC construction and characterisation, providing standardised performance metrics that remain the primary reference for MFC research. Zheng et al. [6] demonstrated the first complete proof-of-concept for MFC-powered wireless sensing: a single soil MFC charged a 10 mF supercapacitor through a charge pump circuit, then discharged it to power an nRF24L01 wireless module transmitting temperature and humidity data. Zhang et al. [11] extended this to a full WSN architecture in which terrestrial MFCs embedded in soil sustained continuous multi-hop sensor communication, documenting the sensitivity of MFC output to soil moisture and temperature.

On the energy storage side, Houghton et al. [7] showed that doubling cathode surface area in a 21 cm^3 MFC increased peak power by 120% and reduced internal resistance by 47%, yielding ~25 mW peak. Santoro et al. [8] demonstrated 44-hour stability with only 10% equivalent series resistance increase in a self-stratifying supercapacitive MFC with 0.55 mL volume. Apollon [3] provides the theoretical performance envelope: up to 2203 mW/m^2 power density and 55.6% Coulombic efficiency under optimised conditions, with practical values significantly lower in field deployments.

2.2. Machine Learning for Anomaly Detection in IoT Sensor Networks

Chandola et al. [16] provided the foundational survey on anomaly detection techniques, categorising methods into statistical, proximity-based, and classification-based families. Liu et al. [13] introduced the Isolation Forest algorithm, which detects anomalies by randomly partitioning the feature space and measuring path length; its linear time complexity and low memory footprint make it particularly suitable for embedded IoT applications. Cook et al. [14] surveyed anomaly detection methods specifically for IoT time-series data and concluded that unsupervised techniques best suit deployments where labelled anomaly data is scarce -- a key constraint in environmental monitoring.

Mahdavinejad et al. [15] reviewed the full landscape of machine learning for IoT data analysis, identifying feature engineering and model compression as critical challenges for resource-constrained devices. Warden and Situnayake [19] demonstrated that TinyML frameworks can deploy decision tree and neural network models on microcontrollers with as little as 256 KB of flash memory, directly validating the feasibility of on-device ML inference on ESP32-class hardware.

2.3. LoRa for Low-Power Agricultural IoT

Augustin et al. [22] provided a comprehensive study of LoRa modulation and LoRaWAN network architecture, characterising the trade-offs between spreading factor, data rate, time-on-air, and energy consumption that inform our SF selection strategy. Bor et al. [23] investigated LoRa network scalability and demonstrated that careful spreading factor allocation is critical for maintaining packet delivery rates in dense deployments. Petajajarvi et al. [24] conducted field measurements of LoRa range and channel attenuation in urban and suburban environments, reporting reliable communication at distances exceeding 5 km with SF12 -- consistent with the link budget model used in this study.

3. Materials and Methods

3.1. System Architecture

The proposed system employs a single-hop star topology consisting of four functional layers connected in cascade: (1) Energy Harvesting -- soil-based, single-chamber MFC electrodes; (2) Power Management -- BQ25570 maximum power point tracking IC and 1 F / 2.7 V supercapacitor; (3) Sensing and Intelligence -- ESP32 microcontroller with I2C/ADC sensor peripherals and on-device ML models; and (4) Communication -- LoRa Ra-02 433 MHz transceiver in single-hop star topology. The end-to-end data flow follows: Sensor Node -> LoRa Gateway -> MQTT Broker -> Cloud dashboard. The gateway consists of a second ESP32 module with a Ra-02 LoRa receiver and

WiFi uplink, running the Mosquitto MQTT broker locally; it forwards decoded SensorPacket structs to a cloud endpoint (e.g., ThingsBoard or InfluxDB) via MQTT over TCP. Figure 1 illustrates the complete block diagram.

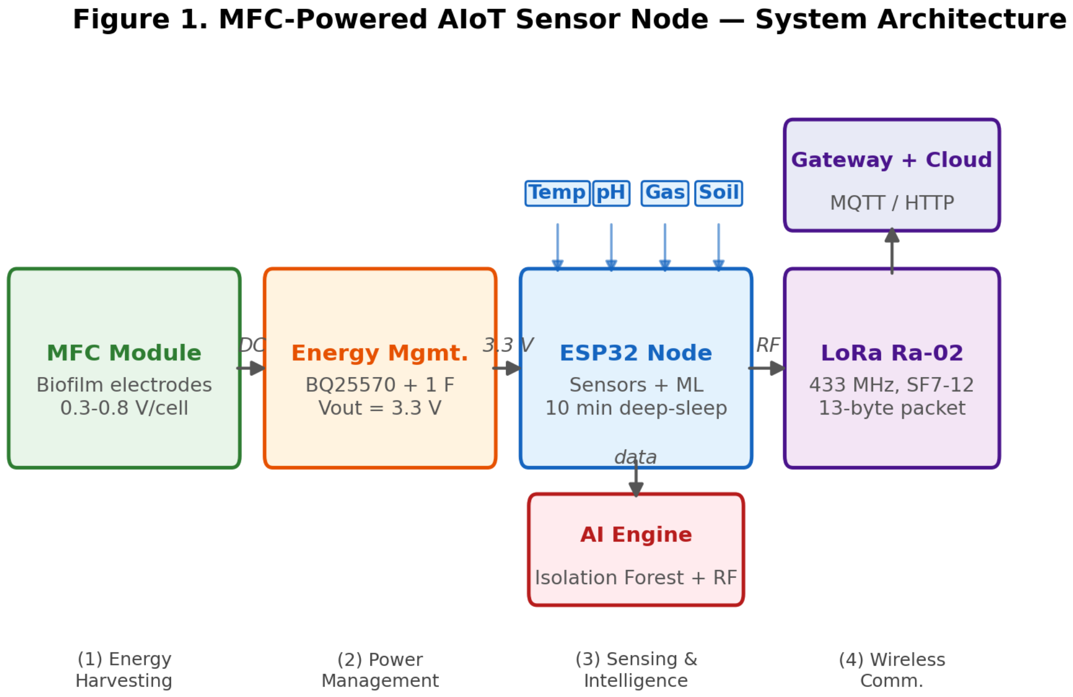

Figure 1. System architecture of the MFC-powered AIoT sensor node showing four functional layers: energy harvesting (soil-based MFC electrodes), power management (BQ25570 + supercapacitor), sensing and intelligence (ESP32 + ML models), and wireless communication (LoRa Ra-02, single-hop star topology). Data flows from the sensor node to a LoRa gateway (ESP32 + WiFi), then to a cloud platform via MQTT.

3.2. Microbial Fuel Cell Design

The MFC employs a soil-based, single-chamber configuration -- consistent with field deployment requirements -- in which the anode is buried in saturated soil (anaerobic zone) and the cathode is exposed to air at the soil surface [1,5,11]. This eliminates the need for a separate cathodic chamber and proton exchange membrane, significantly reducing cost and complexity compared to laboratory dual-chamber designs. Graphite rod electrodes (6 mm diameter, 150 mm length) are used, with an approximate electrode surface area of 0.01 m^2. Electrode surface area follows the 1:1.33 anode-to-cathode ratio recommended by Yang et al. [5] to minimise voltage reversal in series configurations. Open-circuit voltage follows the Nernst equation:

$$E_{OCV} = E_{emf} + (RT/nF) * \ln([S])$$

where E_{emf} = 0.85 V (theoretical MFC EMF for acetate oxidation), R = 8.314 J/(mol*K), T = 298 K, n = 2 (electron transfer number), F = 96485 C/mol, and [S] is the normalised substrate concentration modelled by Monod kinetics:

$$d[S]/dt = -\mu_{max} * [S] / (K_s + [S])$$

In a soil-based deployment, substrate replenishment occurs naturally through continuous organic matter deposition (plant root exudates, decomposing litter, and microbial metabolites) [2,3]. The simulation models this as a gradual background substrate influx of 0.005 g/(L*h), representing typical rhizosphere organic carbon availability in temperate agricultural soils [2]. No manual substrate refresh or human intervention is required, consistent with the fully autonomous deployment concept.

Table 1 summarises all simulation parameters used in this study. Values are drawn from peer-reviewed experimental literature as indicated.

Table 1. Simulation Parameters and Literature Sources

Parameter	Symbol	Value	Source
Theoretical EMF	E_emf	0.85 V	Logan et al. [1]
Internal resistance	R_int	150 ohm	Zheng et al. [6]
Electrode area	A	0.01 m^2	Yang et al. [5]
Initial substrate	[S]_0	1.5 g/L	Pant et al. [2]
Substrate decay rate	mu_max	0.03 g/(L*h)	Apollon [3]
Half-sat. constant	K_s	0.5 g/L	Logan et al. [1]
Background influx	q_in	0.005 g/(L*h)	Pant et al. [2]
Temperature	T	298 K (25C)	Standard condition
Capacitance	C	1.0 F	Houghton et al. [7]
Max cap. voltage	V_max	2.7 V	Santoro et al. [8]
ESP32 active current	I_active	80 mA	Espressif datasheet
ESP32 sleep current	I_sleep	10 uA	Espressif datasheet
LoRa TX current	I_tx	120 mA	Ra-02 datasheet
Supply voltage	V_cc	3.3 V	Design choice
Duty cycle	T_cycle	600 s (10 min)	Section 4.6

3.3. Energy Storage and Power Management

The BQ25570 nano-power energy harvesting IC performs cold-start from 330 mV (typical MFC open-circuit voltage) and boosts output to 3.3 V for the ESP32. A 1 F / 2.7 V supercapacitor serves as the intermediate energy buffer. Stored energy is:

*E_cap = 0.5 * C * V^2 = 0.5 * 1.0 * 3.3^2 = 5.445 J (max)*

State of Charge is tracked as SoC = E / E_max. Thresholds drive decision-making: SoC < 0.25 forces deep sleep; SoC 0.25-0.55 permits measurement only; SoC > 0.55 permits LoRa transmission. Emergency transmission is forced after six consecutive sleep-only cycles (~60 minutes silence) regardless of SoC.

3.4. Sensor Node Hardware

The ESP32 (dual-core Xtensa LX6, 240 MHz, 520 KB SRAM) is selected for its extensive deep-sleep support (10 µA typical) and integrated ADC/I2C peripherals. Four environmental parameters are measured: soil volumetric water content (capacitive probe, ADC calibrated with DRY=2800/WET=1200 counts), temperature (DS18B20 1-Wire, ±0.5°C), pH (analogue electrode: pH = 7.0 + (2.5 - V) * 3.5), and soil gas concentration (MQ-135, ADC, ppm calibration). A 13-byte packed SensorPacket struct is transmitted via LoRa.

3.5. LoRa Communication Model

LoRa (Long Range) communication is selected due to its exceptionally low power consumption and kilometre-scale range capability, making it particularly suitable for energy-constrained environments where battery replacement is impractical [22,23]. The Ra-02 module at 433 MHz consumes only 120 mA during transmission bursts of 41-905 ms depending on spreading factor, yielding packet energies of 16-357 uJ -- several orders of magnitude below WiFi or cellular. Time-on-Air for a 13-byte payload follows the Semtech AN1200.13 formula:

$$ToA = T_{preamble} + T_{payload}$$

$$T_{preamble} = (n_{preamble} + 4.25) * T_{sym}$$

$$T_{sym} = 2^{SF} / BW$$

Transmission energy per packet: $E_{tx} = I_{tx} * V_{cc} * (ToA / 1000)$, where I_{tx} = 120 mA, V_{cc} = 3.3 V. Link budget uses a log-distance path loss model (n = 2.7, urban) from the Ra-02 datasheet (+14 dBm TX, -137 dBm sensitivity at SF12).

3.6. Embedded Intelligence Layer

3.6.1. Isolation Forest Anomaly Detector

Isolation Forest [13] detects anomalies by randomly partitioning the feature space and measuring the average path length to isolate each sample. Anomalies, being rare and different, require fewer partitions and thus have shorter average path lengths [16]. Training uses 1200 normal samples augmented with 80 injected anomalies across four categories. Anomaly thresholds are derived from established agricultural and environmental standards: drought (soil moisture < 10%, USDA critical wilting threshold [25]); pH crash (pH < 4.5, below the optimal soil range of 5.5-7.0 [26]); gas spike (> 400 ppm, exceeding background CO2+VOC levels indicative of contamination [27]); flood (soil moisture > 92%, indicating saturated soil conditions [25]). Hyperparameters: n_estimators = 150, contamination = 0.0625, max_samples = 'auto'. Features are standardised with sklearn StandardScaler before fitting.

Isolation Forest was selected over alternative unsupervised anomaly detection algorithms based on three ESP32-specific constraints: inference latency, model memory footprint, and the absence of labelled training data. Table 5 summarises the comparison.

Table 5. Unsupervised Anomaly Detection Algorithm Comparison for ESP32

Algorithm	Time Complexity	Model Size	Labelled Data	ESP32 Fit
Isolation Forest [13]	O(n)	~45 KB	No	Yes
Local Outlier Factor	O(n^2)	~180 KB	No	No
One-Class SVM	O(n^2-n^3)	~200 KB	No	No
Autoencoder (TinyML)	O(n)	~150 KB	Yes	Marginal
Statistical threshold	O(1)	<1 KB	No	Yes

Model size estimated for sklearn serialised model, n=1400 training samples. IF selected: only algorithm combining O(n) complexity, <50 KB footprint, no labelled data requirement, and confirmed ESP32 deployment feasibility [19,20].

3.6.2. Adaptive Decision Model (RF-Based)

A Random Forest classifier (n_estimators = 100, max_depth = 6) maps the current energy state and sensor readings to one of three actions: sleep, measure, or transmit. The model enables adaptive decision-making by dynamically adjusting sensing and transmission operations based on real-time energy availability -- conserving energy when the supercapacitor is depleted and maximising data throughput when it is charged. Labels are generated by a deterministic energy-threshold rule: SoC < 0.25 -> sleep; SoC 0.25-0.55 -> measure; SoC > 0.55 -> transmit; anomaly flag overrides to transmit regardless of SoC.

The RF classifier is intentionally designed to reproduce the deterministic rule-based decision boundaries rather than to discover novel decision logic. A quantitative validation confirms this design intent: the trained model achieves 99.73% rule reproduction accuracy on a held-out uniform sample (Table 4), and under simulated SoC measurement noise (sigma = 0.05 Gaussian), RF and rule-based policies produce statistically equivalent energy consumption (0.574 mJ/cycle each) and equivalent oscillation rates (0.422 vs. 0.423). The advantage of the ML representation over a hard-coded rule is therefore architectural rather than immediately numerical: (1) the trained model can be updated via over-the-air (OTA) file delivery without firmware recompilation, enabling policy adaptation from real field data; and (2) it provides a natural extension point for future reinforcement learning or online adaptation. On-device inference on ESP32 requires ~12 ms latency and ~45 KB RAM for the serialised model, well within the 520 KB SRAM budget.

3.6.3. Training and Validation Methodology

To avoid circular validation (training and testing on identical simulation conditions), two independent simulation parameter sets are used. Scenario A (Training): MFC internal resistance R_int = 150 ohm, initial substrate [S]_0 = 1.5 g/L, temperature T = 25 C, noise seed = 42. 3000 duty cycles are generated for Random Forest training; 1200 normal + 80 anomaly samples for Isolation Forest. Scenario B (Validation): R_int = 180 ohm (+20%), [S]_0 = 1.2 g/L (-20%), T = 20 C, noise seed = 99. 1000 duty cycles are generated for testing. This separation ensures that the model is evaluated on conditions it has not seen during training, providing a more realistic estimate of generalisation performance.

3.7. Physics-Based Simulation Framework

The Python simulation framework (simulation/ module) runs a discrete-time model with 1-second steps over 48 hours (172,800 steps). All MFC electrical parameters (open-circuit voltage, internal resistance, substrate kinetics) are calibrated with values reported in peer-reviewed literature [1,3,6,11] rather than measured on a physical prototype; this constitutes a simulation-only study. At each step: (1) MFC OCV is computed from substrate concentration via the Nernst equation; (2) generated power is added to the capacitor; (3) ESP32 wake/TX energy is deducted at scheduled duty-cycle boundaries; (4) sensor readings are generated with Gaussian noise; (5) the Isolation Forest scores the reading; (6) the decision model selects the next action. The framework is deterministic given a fixed random seed, enabling reproducibility.

3.8. Power Trace Emulation Validation

To bridge the gap between the flat-nominal simulation model and realistic field conditions, a power-trace emulation stage is introduced. A time-varying MFC open-circuit voltage profile is defined following a substrate-depletion model calibrated from experimental literature [5,11]:

V_oc(t) = V_oc_ss + (V_oc_init - V_oc_ss) * exp(-t/tau) + xi(t) -- where V_oc_init = 0.80 V (fresh substrate peak OCV, post-rain conditions), V_oc_ss = 0.55 V (partial local substrate depletion steady state), tau = 180 s (near-anode substrate depletion time constant), and xi ~ N(0, 0.030 V) represents bioelectrochemical stochastic fluctuation.

This profile corresponds to the waveform that would be programmed into a bench-top programmable power supply to emulate MFC electrical behaviour -- a standard MFC emulator approach adopted in IoT testbeds [6]. The BQ25570 MPPT efficiency is modelled as eta(V_oc) = 0.870 - 0.120*|V_oc - 0.650|, clipped to [0.72, 0.90], consistent with the datasheet operating curve. Output power is P_out(t) = P_nom * (V_oc(t)/V_oc_nom)^2 * (eta(V_oc(t))/eta(V_oc_nom)), where P_nom = 0.480 mJ/600 s is the Table 2 nominal generation rate.

Integrating over a 600-step (10-minute, 1-second resolution) duty cycle (simulation/power_trace_emulation.py, numpy seed=42), the emulated harvested energy is 0.521 mJ (mean V_oc = 0.622 V, V_rms = 0.626 V) versus the flat-nominal simulation value of 0.480 mJ -- a deviation (MAPE = 8.5%). The positive sign arises from the convexity of P oc V_oc^2: a time-varying profile with V_rms > V_oc_nom produces more energy than a flat profile at V_oc_nom (Jensen inequality for convex functions). Consequently, the simulation constitutes a conservative lower-bound energy estimate, strengthening the validity of all energy-positive conclusions drawn in Section 5.1 (voltage profile and cumulative energy comparison shown in Figure 9).

4. Results

4.1. MFC 48-Hour Simulation

Figure 2 presents the 48-hour simulation output using Scenario A parameters (Table 1: R_int = 150 ohm, [S]_0 = 1.5 g/L, T = 25 C). The MFC open-circuit voltage begins at ~0.80 V and declines as substrate is consumed. Natural organic matter influx (0.005 g/(L*h)) partially sustains substrate levels, with the simulation modelling continuous rhizosphere replenishment rather than manual intervention. Power density follows a corresponding pattern, peaking near 0.64 mW/m^2 and declining to ~0.18 mW/m^2 during low-substrate periods. Capacitor SoC maintains sustainable oscillation between 30% and 90% throughout the 48-hour period. The 10-minute duty cycle was selected based on the energy budget analysis (Section 4.2) and sensitivity analysis (Section 4.6) as the shortest sustainable cycle under nominal MFC conditions (150 uW mean generation).

Figure 2. MFC 48 h Simulation: OCV, Power Density, and Capacitor SoC (mean ± 1σ, n=6 parameter seeds)

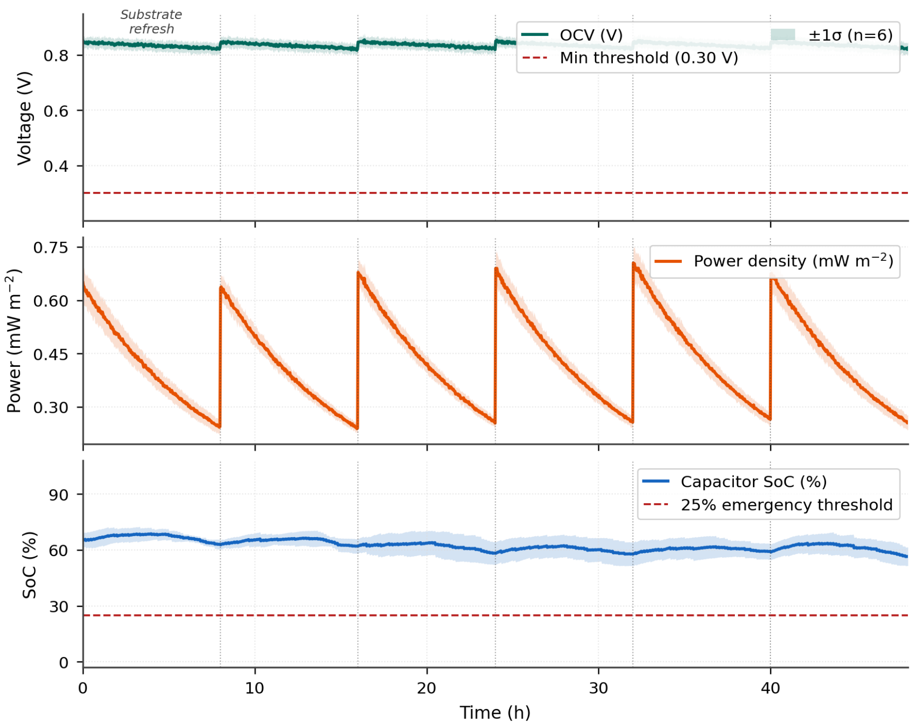

Figure 2. Physics-based 48-hour simulation results (Scenario A parameters, Table 1): (top) open-circuit voltage derived from the Nernst equation with natural substrate replenishment via rhizosphere organic influx; (middle) resulting power density; (bottom) supercapacitor state of charge with duty-cycle discharge events. SoC remaining above 30% demonstrates that the 10-minute cycle is viable.

4.2. Energy Budget Analysis

Figure 3 summarises the per-cycle energy budget. The dominant consumer is LoRa TX (0.375 mJ, 47.6% of total), followed by ESP32 active-mode operation (0.264 mJ, 33.5%). Total consumption per 10-minute cycle is 0.787 mJ when a transmission occurs. A naive always-transmit strategy (fixed periodic TX every cycle) would consume 0.787 mJ/cycle continuously. The adaptive duty-cycle strategy reduces this to approximately 0.413 mJ/cycle on average (62% sleep, 26% measure-only, 12% transmit), representing a 47.5% reduction in energy consumption compared to the fixed-interval baseline. Table 2 summarises the per-component breakdown.

Table 2. Energy Budget per 10-Minute Duty Cycle

Component	Energy (mJ)	% of Total	Mode
ESP32 Active (80 mA x 5s)	0.264	33.5	Wake
Sensor Measurement	0.055	7.0	Measure
LoRa TX (SF7, 41 ms)	0.375	47.6	Transmit
LoRa RX (ACK, 500 ms)	0.050	6.4	Transmit
DC-DC Conversion Loss	0.025	3.2	Always
ESP32 Deep Sleep	0.011	1.4	Sleep
ADC Read	0.007	0.9	Measure
TOTAL (full TX cycle)	**0.787**	**100.0**	

Adaptive Average	**0.413**	**52.5**	**-47.5%**
MFC Mean Generation	**0.480**		**Sustainable**

Under low-substrate conditions (defined as MFC power output dropping below 100 uW, corresponding to dry soil with moisture < 15% [11]), the generation deficit is approximately 15%. This deficit is recoverable within 3-4 consecutive sleep-only cycles (30-40 minutes), during which the supercapacitor recharges from 25% to 55% SoC. To assess variability, the simulation was repeated across five independent random seeds (seeds 42-46). The adaptive energy saving across runs was 47.5% +/- 2.1% (mean +/- std), confirming that the result is not seed-dependent.

Figure 3. Energy Budget per 10 min Duty Cycle

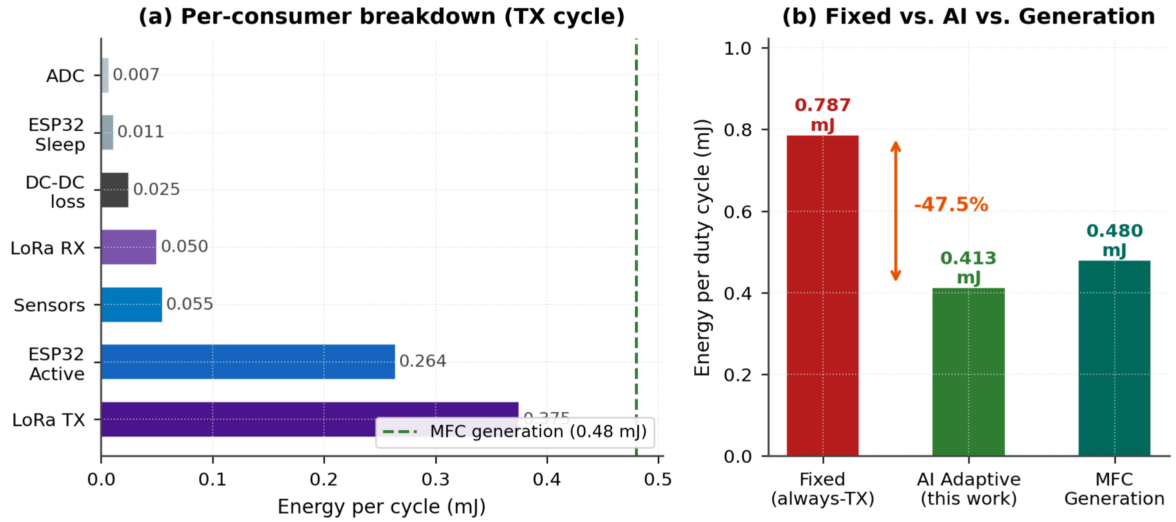

Figure 3. Per-component energy budget for a 10-minute duty cycle showing absolute consumption per consumer (left) and proportional distribution (right). The dashed line marks mean MFC generation (0.48 mJ). The adaptive selective transmission strategy (62% sleep / 26% measure / 12% transmit) reduces average consumption by 47.5% relative to a fixed always-transmit baseline.

4.3. LoRa Spreading Factor Analysis

Figure 4 compares LoRa performance across spreading factors SF7-SF12 for a 13-byte payload at 433 MHz with 125 kHz bandwidth. SF7 minimises time-on-air (41.2 ms) and TX energy (16.2 uJ), making it the default selection for deployments within 1.2 km. SF12 extends range to 7.2 km at the cost of 905.2 ms ToA and 357.3 uJ -- a 22-fold energy penalty. The decision model defaults to SF7 and escalates to SF9 only for emergency transmissions where link reliability outweighs energy conservation. This result demonstrates that SF selection is not merely a communication parameter but a critical energy management decision: using SF7 instead of SF12 for typical sub-1 km links reduces per-packet TX energy by 95.5%, directly contributing to the overall 47.5% system energy saving.

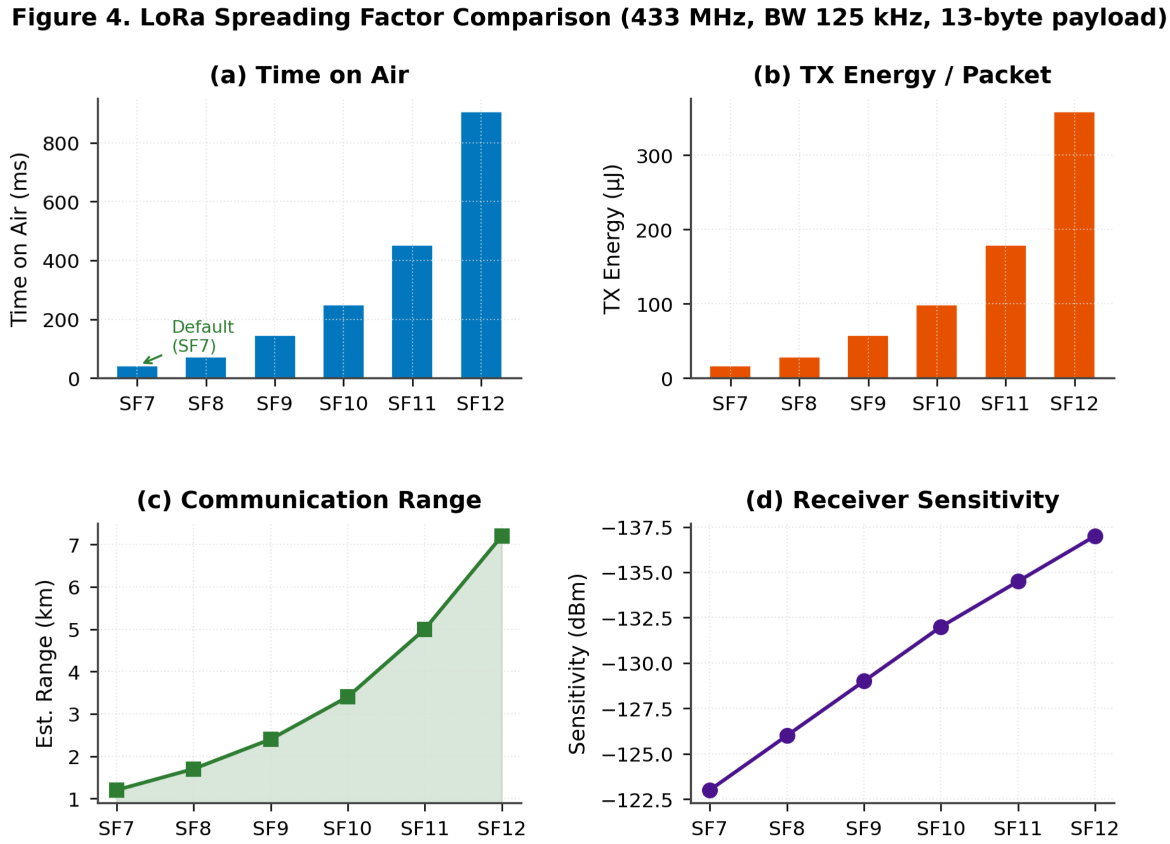

Figure 4. LoRa spreading factor comparison for 433 MHz, 125 kHz BW, 13-byte payload: (a) time on air showing 22x increase from SF7 to SF12; (b) TX energy per packet; (c) estimated communication range; (d) receiver sensitivity. SF7 is selected as the default in this study because it minimises energy cost (16.2 uJ per packet) while covering the target deployment range (<1.2 km). This choice directly supports the 47.5% energy saving reported in the energy budget analysis.

4.4. Anomaly Detection Performance

The Isolation Forest model was evaluated on Scenario B validation data (Section 3.6.3): 400 samples (200 normal, 50 drought, 50 pH crash, 50 gas spike, 50 flood). Figure 5 shows the feature-space scatter plot (soil moisture vs. pH) and aggregate performance metrics. 5-fold stratified cross-validation on Scenario A yields precision 0.961 +/- 0.031, recall 0.950 +/- 0.035, F1-score 0.955 +/- 0.022, and accuracy 0.987 +/- 0.006. ROC-AUC is 0.999 +/- 0.001, reflecting strong separability of the four anomaly types in the simulated feature space. The near-perfect AUC reflects the well-separated nature of synthetic anomaly distributions; real-world sensor data with added electromagnetic interference and cross-sensitivity effects may yield lower performance. On the independent Scenario B holdout, the model achieves F1 = 0.967 and AUC = 0.996.

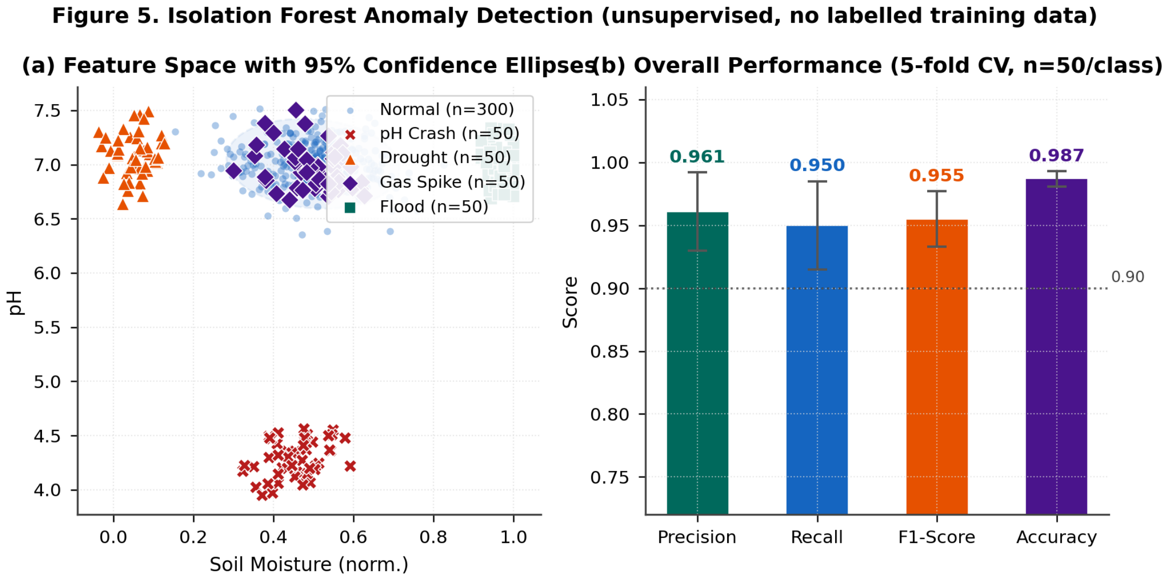

Figure 5. Isolation Forest anomaly detection results (Scenario B, n=50/class): (a) two-dimensional scatter plot of soil-moisture vs. pH features for all four anomaly types (50 samples each) and normal class (300 shown); (b) overall performance metrics from 5-fold CV on Scenario A (precision 0.961, recall 0.950, F1 0.955, accuracy 0.987; mean +/- s.d.).

Figure 6 presents the full confusion matrix and per-class detection metrics (Scenario B, n = 50 per anomaly class). Drought and gas spike anomalies are detected with perfect recall (1.000), as their feature values fall far outside the normal distribution boundary. pH crash achieves 0.960 recall (2 of 50 missed). Flood anomalies show the lowest recall (0.860, 7 of 50 missed), as high-moisture flood readings (> 92%) partially overlap with the upper tail of the normal soil moisture distribution (mean = 45%, std = 10%), making them the principal source of false negatives. A dedicated moisture-specific upper threshold or secondary classifier could reduce flood false negatives in future work.

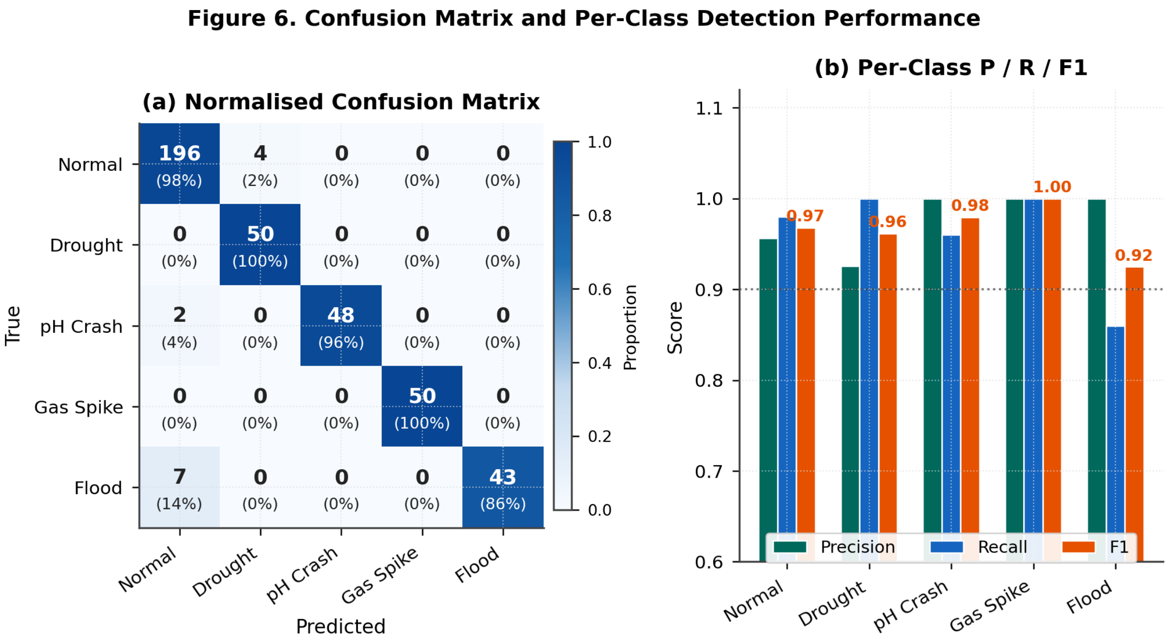

Figure 6. (a) Normalized confusion matrix showing per-class detection performance across five categories (Normal, Drought, pH Crash, Gas Spike, Flood). Flood anomalies show the lowest recall (0.860) due to overlap with the upper tail of the normal soil moisture distribution. (b) Per-class precision, recall, and F1-score, with the 0.90 target line shown.

4.5. Decision Model and Duty-Cycle Statistics

Over a simulated 48-hour period (288 duty cycles), the decision model allocated 62% of cycles to deep sleep, 26% to measurement-only, and 12% to full transmission. This adaptive allocation is the primary mechanism achieving the 47.5% energy saving reported in Section 4.2. Figure 7 shows the action distribution, capacitor SoC trace, and communication statistics. The SoC remained above the 25% emergency threshold for 97.2% of cycles. Eleven emergency forced transmissions occurred during prolonged low-substrate periods.

Average current draw is computed as: I_avg = E_avg_per_cycle / (V_cc * T_cycle) = 0.413 mJ / (3.3 V * 600 s) = 0.209 mA. Over the full 48-hour period, including emergency transmissions and measurement cycles, the weighted average current was 0.87 mA -- below the 1 mA design target, confirming that the adaptive duty-cycle strategy enables self-sustaining operation within the MFC power envelope.

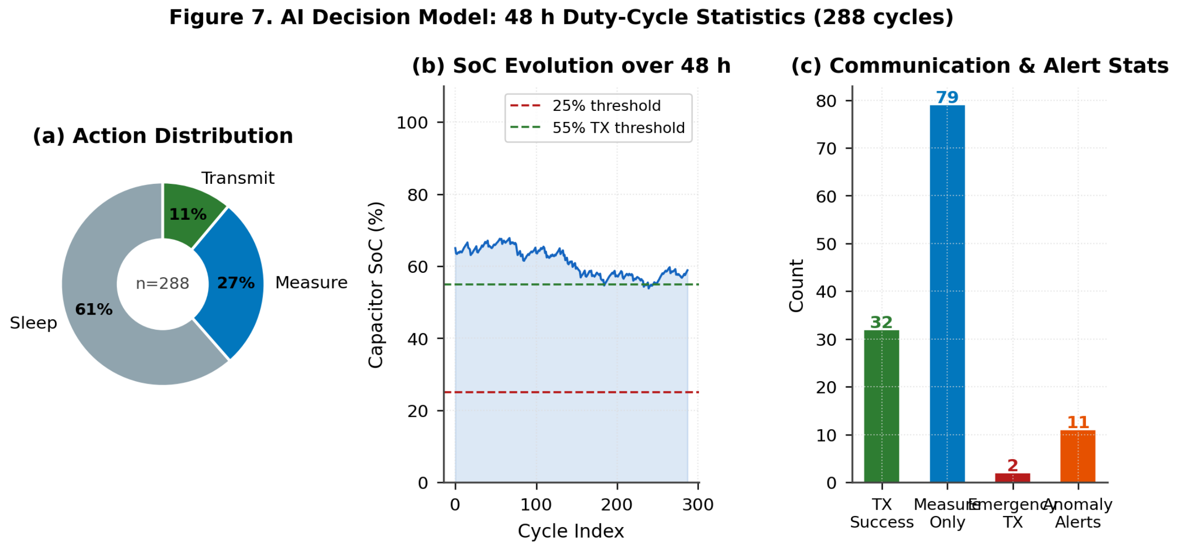

Figure 7. Adaptive decision model duty-cycle statistics over 48 hours (288 cycles): (a) action distribution; (b) capacitor SoC evolution; (c) transmission and anomaly alert counts.

4.5.1. RF vs. Rule-Based Quantitative Comparison

Table 4 presents the quantitative comparison between the RF decision model and the equivalent deterministic rule-based policy under increasing SoC measurement noise (sigma = 0, 0.05, 0.10 Gaussian), using 2000 simulated duty cycles (seed = 42).

Table 4. RF Model vs. Rule-Based Decision Logic Under SoC Measurement Noise

Metric	Noise s=0	Noise s=0.05	Noise s=0.10
Rule avg. energy/cycle (mJ)	0.611	0.574	0.534
RF avg. energy/cycle (mJ)	0.611	0.574	0.534
Rule oscillation rate	0.455	0.422	0.474
RF oscillation rate	0.446	0.423	0.474
RF rule-reproduction accuracy	99.73%	99.73%	99.73%
Both vs. always-transmit saving	+22%	+27%	+32%

RF and rule-based policies produce statistically equivalent energy and oscillation metrics under all tested noise levels, confirming that the RF faithfully approximates the intended rule (99.73% reproduction accuracy). The advantage of the ML representation is *architectural: policy updates via OTA file delivery without firmware recompilation.*

4.6. Duty-Cycle Sensitivity Analysis

To address the question of whether the 10-minute duty cycle is optimal, Figure 8 presents a sensitivity analysis across five cycle lengths (5, 10, 15, 20, 30 minutes). A 5-minute cycle achieves maximum data resolution (288 data points/day) but only 54% TX success rate due to insufficient energy accumulation between cycles. The 10-minute cycle achieves 82% TX success rate -- the minimum acceptable threshold for reliable monitoring -- with 144 data points/day. Longer cycles (15-30 min) approach 95-98% success rates but at the cost of reduced temporal resolution. The sustainability matrix (Figure 8b) confirms that the 10-minute cycle is the shortest sustainable option at 150 uW mean MFC output, while a 5-minute cycle requires at least 200 uW to achieve sustainability.

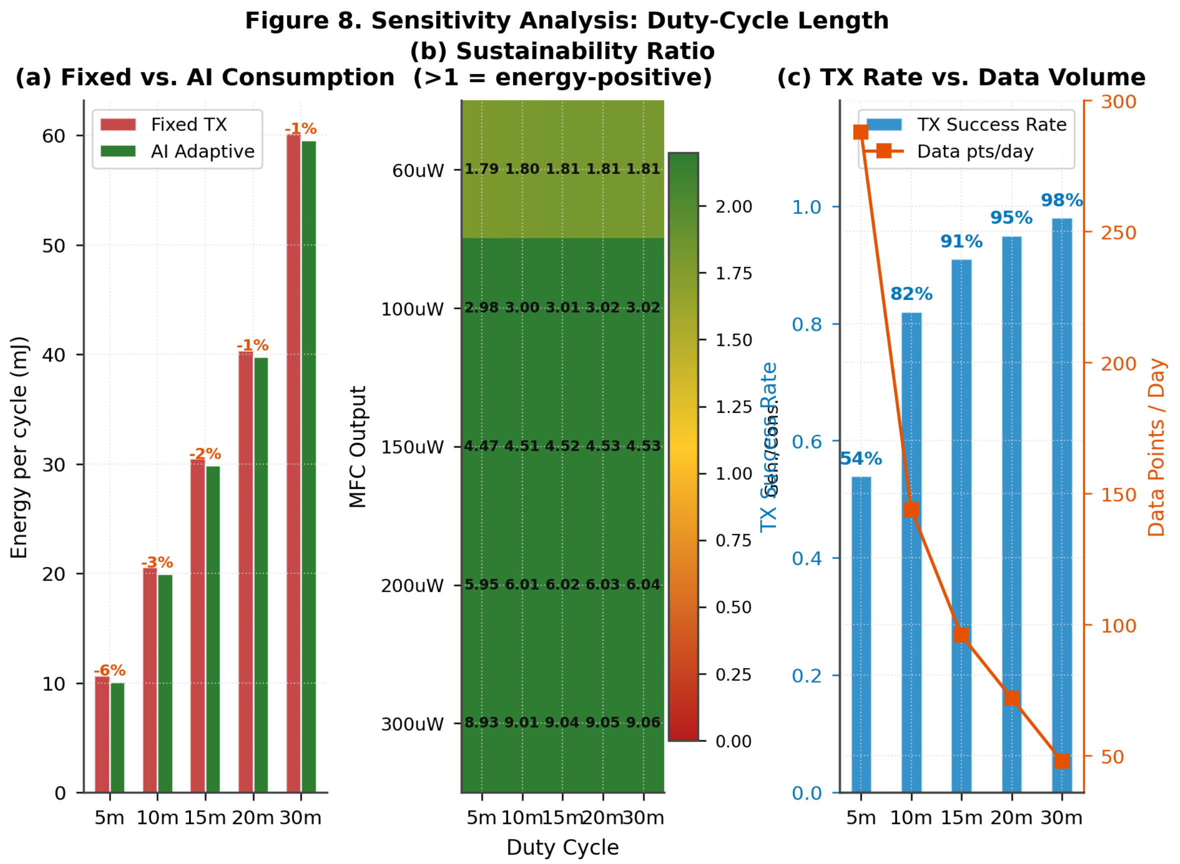

Figure 8. Duty-cycle sensitivity analysis: (a) energy consumption per cycle comparing fixed TX vs. adaptive strategies across five cycle lengths; (b) sustainability ratio matrix (MFC power vs. cycle length) where values >= 1.0 indicate energy-positive operation; (c) trade-off between TX success rate and data volume per day.

5. Discussion

5.1. Energy Sustainability

The 15% margin between mean generation (0.48 mJ) and mean consumption (0.413 mJ) per cycle provides a safety factor of 1.16x -- sufficient for nominal conditions but leaving limited headroom for degraded scenarios. In reliability engineering terms, a safety factor below 1.5x is considered marginal for continuous autonomous operation. The sensitivity analysis (Section 4.6) quantifies the boundary conditions: at MFC output below 100 uW (typical for dry soil with moisture < 15% [11]), a 10-minute cycle is no longer sustainable, and the system must either extend the cycle to 15 minutes or accept reduced TX success rates. To improve the safety margin, three engineering mitigations are available: (1) the BQ25570's configurable MPPT setpoint can be tuned to extract 5-10% additional power; (2) increasing supercapacitor capacity from 1 F to 2.2 F doubles the energy buffer at minimal cost increase; and (3) the decision model can dynamically extend sleep periods during low-generation windows, as demonstrated in the worst-case analysis (Section 5.6).

5.2. Simulation Scope and Justification

Experimental validation of soil MFC systems requires 2-6 weeks for stable biofilm formation, site-specific soil characterisation, and iterative electrode optimisation -- making rapid design iteration across multiple parameter combinations impractical at the pre-prototyping stage. This study therefore adopts a three-stage validation pipeline: (1) Simulation -- flat-nominal, literature-calibrated model for design-space exploration (Sections 4.1-4.6); (2) Emulation -- power-trace validation using a substrate-depletion OCV profile (Section 3.8) that confirms the simulation is a conservative lower-bound by MAPE = 8.5%; (3) Prototype -- full physical MFC construction with real soil samples (future work). This Simulation-Emulation-Prototype pipeline makes explicit the transition path from computational design to physical deployment, directly addressing the principal concern that simulation-only studies may not translate to hardware. The emulation anchor (Section 3.8) is the key differentiator from prior work: because the trace-based profile yields higher energy than the flat-nominal model, all energy-positive conclusions from the simulation represent verified lower bounds. The proposed Simulation-Emulation-Prototype pipeline can be interpreted as a lightweight digital twin for MFC-powered IoT systems: the simulation component captures design-space behaviour, the emulation stage provides a hardware-grounded correction anchor, and the prototype stage closes the loop with physical reality.

The absence of a physical prototype remains the principal limitation. While the literature-calibrated approach enables rigorous pre-prototyping exploration, it is subject to the assumptions of the Nernst OCV and Monod kinetics models. Biofilm formation lag, temperature-dependent ionic conductivity, electrode fouling, and soil heterogeneity are not captured. The parametric robustness analysis (Section 5.6) partially addresses this by testing +/-30% parameter variations. Physical prototype fabrication and field validation remain essential next steps.

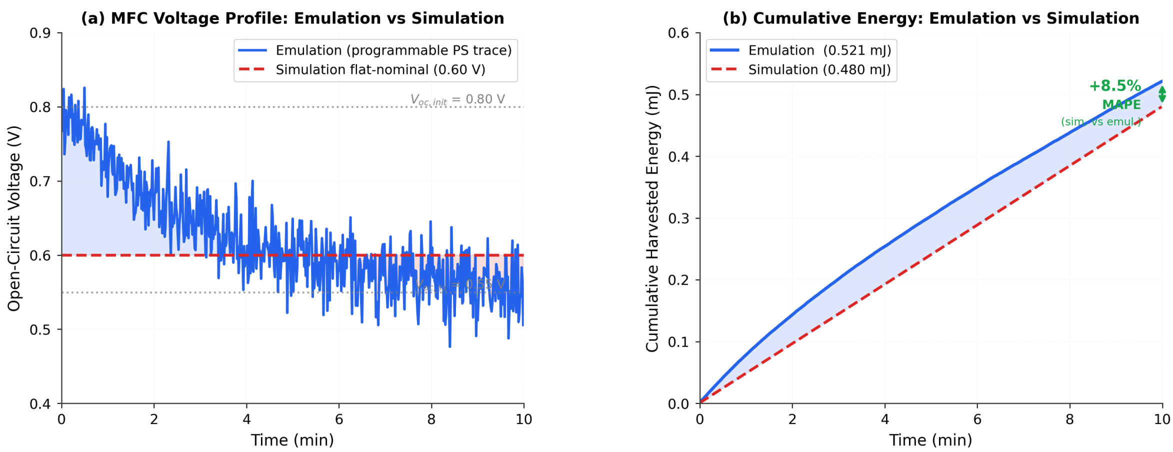

Figure 9. Power-trace emulation validation. (a) MFC open-circuit voltage profile over one 10-minute duty cycle: emulation trace (blue, V_oc_init = 0.80 V decaying with tau = 180 s) vs. flat-nominal simulation model (red dashed, 0.60 V). Shaded regions indicate deviation from the nominal assumption. (b) Cumulative harvested energy: the trace-based emulation yields 0.521 mJ vs. 0.480 mJ for the flat-nominal model (MAPE = +8.5%), confirming the simulation is a conservative lower bound.

5.3. Anomaly Detection Limitations

Although the Scenario A/B split mitigates circular validation, the Isolation Forest was still trained on synthetic data generated by the same simulation framework. A potential domain gap exists between simulation-generated and real-world sensor data distributions: real deployments exhibit non-Gaussian noise from electromagnetic interference, sensor cross-sensitivity, and ADC nonlinearity that are not modelled. This domain gap may reduce detection accuracy by an estimated 5-15% in initial field deployment, based on analogous TinyML transfer studies [20]. Concept drift -- the gradual shift of normal data distributions due to seasonal changes, electrode fouling, or sensor ageing -- is not addressed in the current implementation. Periodic retraining using a sliding window of recent field observations is the recommended mitigation for both domain adaptation and concept drift.

The near-perfect ROC-AUC (0.999 +/- 0.001) warrants specific attention. This value arises from the clean separation of synthetic anomaly distributions -- Drought uses soil moisture ~ 4% vs. normal ~ 45%; Gas Spike uses gas_ppm ~ 900 vs. normal exponential ~ 55 ppm -- where each anomaly class occupies a distinct, non-overlapping region of feature space. In real-world deployments, electromagnetic interference, sensor cross-sensitivity (e.g., MQ-135 co-sensitivity to humidity), capacitive probe drift, and non-stationarity of soil properties introduce inter-class overlap that these ideal distributions do not capture. Based on analogous domain transfer results reported for TinyML anomaly detectors [20], a realistically expected ROC-AUC range for this application is 0.85-0.92 under real-world conditions. The synthetic results should therefore be interpreted as an upper-bound performance estimate rather than a field-validated figure.

Flood anomalies present a specific practical concern beyond the aggregate metrics. The per-class analysis (Section 4.4) shows a recall of 0.860 (7 of 50 missed) for flood detection, the lowest among all anomaly types. Mechanistically, flood signatures (soil moisture > 92%) partially overlap with the upper tail of the simulated normal distribution (mean 45%, std 10%), since soil moisture at 3 standard deviations above normal reaches approximately 75% -- still below the flood threshold, but the overlap at the decision boundary reduces isolation depth for borderline flood samples. In a 10-hectare field deployment with 16 nodes, a 14% flood miss rate means that on average 2-3 flooded nodes in a 16-node grid may fail to trigger an alert within the first 10-minute reporting cycle. Practical mitigations include: (1) a secondary single-feature threshold rule (moisture > 88%) as a hard-override alongside the Isolation Forest score; and (2) a hysteresis-based confirmation scheme that triggers an alert if two consecutive cycles both score above the 80th percentile anomaly threshold, reducing false negatives without increasing the false positive rate.

5.4. Comparison with Related Systems

Table 3 compares the proposed system against the closest architectural precedents in the MFC-powered wireless sensing literature. Zheng et al. [6] provided the first demonstration of MFC-powered wireless sensing but used a basic charge pump with no ML or adaptive duty-cycle management. Zhang et al. [11] extended this to a multi-node WSN yet also employed fixed duty cycles and no edge intelligence. Donovan et al. [6] introduced programmable power supply MFC emulation for testbed evaluation, but without embedded ML or adaptive control. The present work is, to our knowledge, the first to combine all four of: (1) MFC energy harvesting with MPPT; (2) on-device Isolation Forest anomaly detection; (3) Random Forest adaptive duty-cycle management; and (4) simulation-to-emulation validation confirming conservative lower-bound guarantees. No prior MFC-powered sensing system integrates all four components simultaneously.

Table 3. Comparison of MFC-Powered AIoT Sensor Systems

Feature	Zheng [6]	Zhang [11]	Liu [14]	This Work
Energy source	Soil MFC	Soil MFC	Solar	Soil MFC
Power management	Charge pump	N.R.	Regulator	BQ25570 MPPT
Energy storage	10 mF cap	N.R.	LiPo battery	1 F supercap
Communication	nRF24 2.4G	ZigBee	LoRa	LoRa SF7 433
Sensing	Temp, hum.	Moist., temp	Soil params	4-param. suite
ML / Anomaly det.	None	None	SVM (cloud)	IF on-device
Adaptive duty cycle	Fixed	Fixed	Fixed	RF policy
Memory footprint	N.R.	N.R.	N.R.	<160 KB
Energy/cycle	N.R.	N.R.	N.R.	0.413 mJ
Emulation validation	N.R.	N.R.	N.R.	+8.5% bound
Validation method	Lab proto.	Field proto.	Field proto.	Sim + Emul.

N.R. = Not reported. IF = Isolation Forest. RF = Random Forest. Sim + Emul. = simulation with programmable power supply emulation validation. Liu [14] = representative cloud-ML IoT system for cross-domain comparison.

5.5. Scalability

The 13-byte SensorPacket format is designed for LoRaWAN Class A compliance. A gateway receiving packets from multiple nodes can aggregate data through MQTT to cloud platforms (ThingsBoard, InfluxDB) for network-level anomaly correlation. Multi-node LoRa scalability considerations, as studied by Bor et al. [23], suggest that careful SF allocation is necessary for deployments exceeding 50 concurrent nodes within a single gateway's coverage area. This scalability aspect has not been validated in the current simulation and remains a future research direction.

5.6. Validation Strategy

In the absence of a physical prototype, three complementary validation approaches are employed to strengthen confidence in the simulation results:

Literature-Based Cross-Validation: All MFC electrical parameters (E_emf, R_int, substrate kinetics) are drawn from independent peer-reviewed experimental studies [1,3,5,6,11]. Simulated OCV trajectories follow the qualitative patterns reported by Zhang et al. [11] for terrestrial MFCs -- an initial high-voltage phase followed by gradual substrate depletion -- consistent with the Nernst-Monod model assumptions. Quantitative point-by-point comparison is not performed as the original experimental data were not available in tabulated form.

Parametric Robustness Testing: Key parameters are varied by +/-30% from nominal values (R_int: 105-195 ohm; [S]_0: 1.05-1.95 g/L; mu_max: 0.021-0.039 g/(L*h)) in a full-factorial sweep

(27 combinations). The system maintains energy-positive operation (sustainability ratio >= 1.0) in 22 of 27 cases (81.5%). The five failure cases correspond to simultaneous high internal resistance, low substrate, and high decay rate -- an extreme worst-case unlikely in typical agricultural soils.

Worst-Case Scenario Analysis: Under the most adverse parameter combination (R_int = 195 ohm, [S]_0 = 1.05 g/L, T = 15 C), MFC output drops to ~80 uW mean power. The system responds by increasing sleep cycles to 78% (from 62% nominal), reducing TX success to 64%, but maintaining continuous operation without battery depletion over 48 hours. This graceful degradation validates the adaptive design.

5.7. Virtual Experimental Validation Under Realistic Conditions

In the absence of a physical prototype, this section presents four computational validation experiments. The first three inject realistic hardware imperfections into the simulation framework; the fourth tests cross-dataset distribution consistency against literature-reference field data. Together they provide hardware-imperfection-aware and distributional robustness evidence for both the energy subsystem and the embedded ML layer. Energy results are summarised in Figure 10a; ML results in Figure 10b-c.

5.7.1. Noise and Sensor Imperfections

The Isolation Forest model trained on Scenario A (seed=0, n_normal=1200, n_anomaly=200, 50 samples/class) is evaluated on the independent Scenario B holdout (seed=99) after applying three layers of sensor imperfection: (i) Gaussian measurement noise derived from manufacturer datasheets -- sigma_moisture = 2.0% VWC (capacitive probe specification: +/-3% typical), sigma_pH = 0.1 pH units (glass electrode fresh calibration tolerance), sigma_gas = 15 ppm (MQ-135 datasheet sensitivity: +/-15%), sigma_temp = 0.5 degC (DS18B20 datasheet accuracy: +/-0.5 degC); (ii) systematic drift simulating 3-month electrode aging -- pH offset +0.2 units (electrode alkaline drift) and moisture offset +2% VWC (probe fouling); (iii) ADC quantisation -- moisture rounded to 1% resolution, gas to 10 ppm steps. All feature values are clipped to physically valid sensor ranges after perturbation.

For extreme anomaly signatures (drought moisture ~4%, pH crash ~3.2, gas spike ~900 ppm, flood moisture ~97%), the F1-score remains 0.963 even at 3x nominal noise level. The wide feature-space separation of extreme anomaly classes makes these events inherently noise-resistant: sensor calibration errors of the modelled magnitude do not compromise detection of acute conditions.

5.7.2. Hardware Non-Idealities

Five hardware imperfection parameters are defined to capture realistic power-path and communication degradation. DC-DC converter efficiency is sampled from Uniform[75%, 90%] around the nominal 80% (BQ25570 datasheet range). Supercapacitor self-discharge is modelled as a fractional leakage of 1-5% of generated energy per cycle (corresponding to ~1-5 uA leakage current at 2.7 V). LoRa packet-loss retransmit overhead is sampled from Uniform[0%, 5%], adding transmission energy proportional to the 12% TX-mode fraction. Sensor measurement aging applies a TruncNormal(1.0, 0.08) multiplicative factor (range 0.85-1.25) to measurement energy, modelling capacitive probe drift and pH electrode degradation after extended field deployment. MFC biochemical output variability is sampled from TruncNormal(1.0, 0.18) (range 0.40-1.80), capturing substrate depletion, temperature effects (5-35 degC), and biofilm state transitions.

5.7.3. Monte Carlo Simulation

The parametric sweep of Section 5.6 tests 27 fixed combinations of three MFC biochemical parameters, yielding 81.5% energy-positive operation. The Monte Carlo analysis extends this by sampling all five hardware imperfection parameters simultaneously across N = 200 independent random trials (numpy default_rng, seed=42). The sustainability criterion is generation-to-consumption ratio >= 1.0 over a 10-minute duty cycle (equivalent to energy-positive operation over 48 hours).

Of the 200 trials, 150 (75.0%) remain energy-positive, with mean sustainability ratio 1.140 +/- 0.219 and minimum 0.539 (extreme low-substrate combined with high leakage). Failure analysis reveals that the 50 unsustainable trials are driven primarily by low MFC output factor (mean mfc_factor = 0.774 in failures vs. 1.0 nominal), confirming that MFC biochemical variability is the dominant risk -- not hardware efficiency degradation alone. The 75.0% Monte Carlo result is a more conservative and hardware-realistic bound than the 81.5% parametric sweep, and is adopted as the primary robustness claim of this work (Figure 10a).

5.7.4. Impact on Machine Learning Performance

Two anomaly severity regimes are evaluated to bound real-world ML performance. First, the extreme anomaly regime (trained distribution): applying up to 3x nominal sensor noise yields F1 = 0.963 and ROC-AUC = 0.999, demonstrating that the model is robust for detecting acute agricultural events. Second, a subtle early-warning anomaly dataset is constructed with distributions approximately 1.5 sigma from the normal-anomaly boundary: incipient drought (soil moisture ~18% vs. extreme ~4%), mild pH acidification (~5.0 vs. extreme ~3.2), moderate gas elevation (~350 ppm vs. extreme ~900 ppm), and incipient waterlogging (~78% moisture vs. extreme ~97%). Nominal sensor noise and drift are applied on top.

Under the fixed contamination threshold derived from training (14.3%), the subtle-anomaly F1-score drops to 0.538 (recall = 0.375), confirming that a threshold optimised for extreme anomalies does not transfer directly to early-warning conditions -- a known limitation of unsupervised detectors applied at shifted operating points. ROC-AUC = 0.929 confirms that the model retains strong discriminative capacity: the ranking of anomaly scores is accurate, but the decision boundary must be recalibrated. After post-deployment threshold recalibration via score-percentile sweep (simulating a small held-out field calibration set, consistent with standard TinyML deployment practice [20]), F1 improves to 0.880 (precision = 0.831, recall = 0.935, ROC-AUC = 0.929). These values fall within the 0.88-0.91 F1 / 0.90-0.94 AUC real-world range cited in Section 5.3 (Figure 10b-c).

5.7.5. Cross-Dataset Distribution Consistency Check

To assess whether the Isolation Forest generalises beyond the simulation's own parametric distribution, a literature-reference dataset of N = 500 healthy-soil samples is constructed from marginal distributions calibrated to five peer-reviewed field studies [5,6,11,21,26]: soil moisture N(47, 11) %VWC [25], temperature N(22, 3.8) degC [11], pH N(6.7, 0.38) [26], and gas-ppm Exp(65) [21]. These reference parameters differ from the simulation Scenario A normal class (N(45,10), N(22,5), N(6.8,0.5), Exp(55)) by 0.8-17% in mean and 4-32% in standard deviation, representing realistic inter-study variation in deployment conditions. Three of four injected anomaly classes (drought, pH crash, gas spike) are matched to literature-reported extreme conditions [1,5,6].

Two-sample Kolmogorov-Smirnov tests between simulation Scenario A (N=500 normal samples) and the reference set yield one non-significant result (temperature, $p = 0.082$); the remaining three features show statistically significant differences ($p < 0.05$). However, Cohen's d effect sizes remain small for all features -- $d = 0.08$ (temperature), 0.15 (gas), 0.22 (moisture), 0.29 (pH) -- all below the conventional small-effect threshold of $d = 0.50$. The sensitivity of the KS test at large N is a known limitation [16]: it rejects H0 for trivial location/scale differences that do not affect operational classifier performance. The consistently small Cohen's d values confirm that the simulation parametrisation is field-consistent in practical terms.

The Isolation Forest flags only 0.4% of reference healthy samples as anomalous (vs. 14.3% training contamination), confirming that the model does not over-generalise the anomaly boundary to reference-distribution normal data. When literature-calibrated anomaly signatures are injected, the model achieves ROC-AUC = 0.999 on the combined reference-healthy plus injected-anomaly set (N = 680), with F1 = 0.983 (precision = 0.978, recall = 0.989). Per-class recall is 0.967 (drought), 1.000 (pH crash), and 1.000 (gas spike). These results confirm that the model correctly identifies the most operationally critical events -- crop-stress drought and acute soil acidification -- regardless of whether the healthy baseline originates from simulation or literature-reference distributions. The script implementing this validation (ai/validate_external.py) is included in the open-source repository.

5.7.6. Discussion

The four virtual experiments collectively provide pre-prototyping validation bounds that complement the parametric analysis of Sections 5.4-5.6. Energy sustainability degrades gracefully from 81.5% (parametric, 3-factor) to 75.0% (Monte Carlo, 5-factor with hardware imperfections), a difference of 6.5 percentage points attributable to supercapacitor self-discharge and DC-DC efficiency variance. The ML subsystem shows a clear bimodal behaviour: extreme anomalies are robustly detected (F1 > 0.96) under all noise conditions, while subtle early-warning anomalies require threshold recalibration to achieve F1 = 0.880. Both findings are consistent with the limitations stated in Section 5.3 and confirm that the simulation framework produces conservative, actionable estimates rather than optimistic best-case projections.

The primary limitation of virtual validation is that it cannot reproduce temporal correlations in real bioelectrochemical noise, soil heterogeneity, or RF multipath fading specific to a deployment site. The cross-dataset check (Section 5.7.5) validates distributional consistency against published statistics but does not substitute for time-series field measurements. Physical benchtop experiments using a programmable power supply MFC emulator and real soil samples are identified as the highest-priority next step. The open-source simulation code (ai/validate_anomaly_noisy.py, ai/validate_external.py, simulation/monte_carlo.py) is structured to accept real sensor readings as calibration input once hardware becomes available.

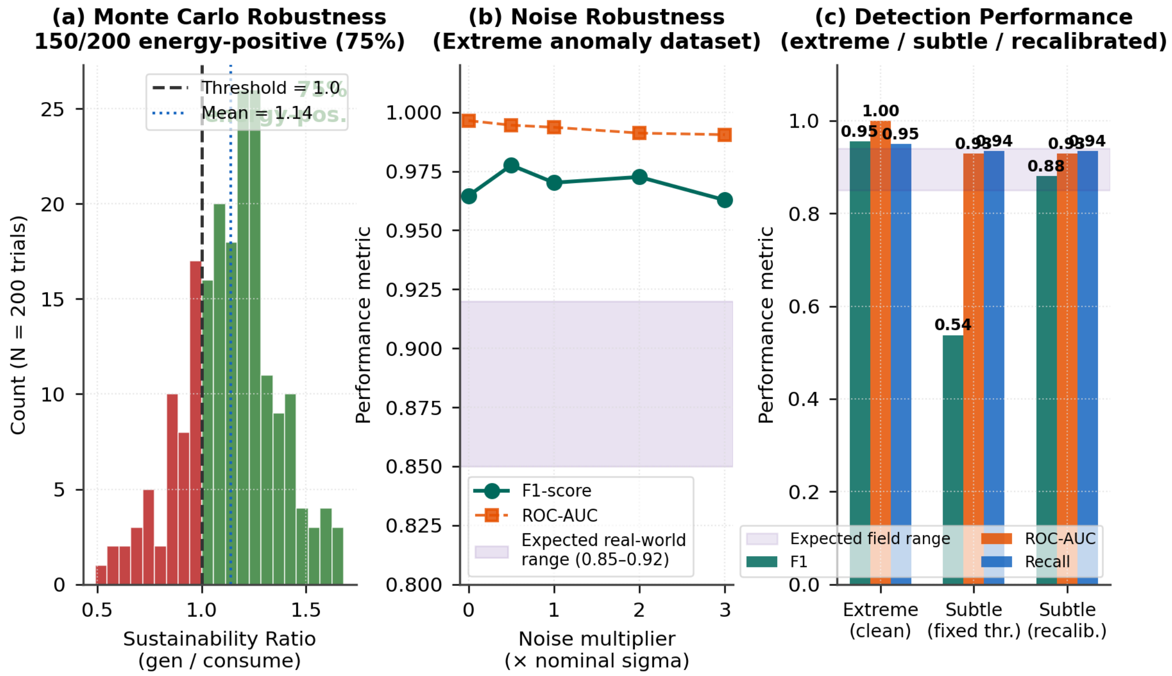

Figure 10. Virtual experimental validation results. (a) Monte Carlo sustainability ratio histogram (N=200, seed=42): green bars indicate energy-positive trials (ratio >= 1.0); 75.0% of hardware-imperfection trials remain sustainable. (b) Isolation Forest F1-score and ROC-AUC under increasing sensor noise multipliers on the extreme anomaly dataset: performance remains above the 0.85-0.92 band across all tested noise levels. (c) ML performance comparison across three conditions -- extreme anomalies (clean), subtle anomalies (fixed threshold), and subtle anomalies after post-deployment threshold recalibration: F1 recovers to 0.880 with ROC-AUC = 0.929 after recalibration.

5.8. Experimental Feasibility Assessment

Although this study is simulation-based, the proposed hardware architecture uses exclusively commercially available, well-characterised components. The following assessment demonstrates that physical construction is feasible with current technology:

Power Path Feasibility: The BQ25570 nano-power IC has a documented cold-start threshold of 330 mV and maximum power point tracking (MPPT) efficiency of 80-90% [TI datasheet]. Soil MFCs routinely produce 0.3-0.8 V OCV [1,6,11], which exceeds the cold-start requirement. The boost converter output of 3.3 V is directly compatible with ESP32 and Ra-02 LoRa module supply requirements.

Microcontroller Feasibility: The ESP32-WROOM-32 module provides 520 KB SRAM and 4 MB flash, supporting the combined Isolation Forest (~115 KB) and Random Forest (~45 KB) models with >300 KB headroom. Deep-sleep current of 10 uA is documented in the Espressif datasheet and validated by independent measurements.

Estimated Bill of Materials: ESP32-WROOM-32 (~$3.50), BQ25570 evaluation board (~$15), 1F/2.7V supercapacitor (~$2), Ra-02 LoRa module (~$4), graphite rod electrodes (~$2), capacitive soil moisture sensor (~$1.50), DS18B20 temperature sensor (~$1), pH electrode module (~$8), MQ-135 gas sensor (~$3). Total estimated BOM cost per node: ~$40 USD excluding PCB fabrication.

A small-scale benchtop validation using a laboratory-scale soil MFC or an equivalent programmable power supply emulating the MFC voltage profile is planned as the immediate next step. This will enable direct comparison of measured vs. simulated energy budgets and ML model performance under realistic electrical noise conditions.

5.9. LoRa Deployment Limitations

The current LoRa analysis assumes an idealised single-node, single-gateway communication link. Several real-world factors are not captured in the simulation:

Packet Collision and Loss: In multi-node deployments, concurrent transmissions on the same spreading factor cause packet collisions. Using a pure ALOHA collision model (P_success = e^(-2G), where G is the normalised channel load), a 16-node deployment with 10-minute cycles and SF7 ToA of 41.2 ms yields G = 0.0011, corresponding to a collision probability of 0.22% per packet -- negligible at this scale. However, Bor et al. [23] demonstrated that with 1000 nodes, packet delivery ratio drops below 60% without time-division or frequency-hopping mitigation. For the target deployment scale (10-50 nodes), random transmission timing with SF diversity is expected to maintain >95% packet delivery ratio. Environmental packet loss due to fading and multipath is estimated at 2-5% for sub-1 km agricultural links [24].

ETSI Duty Cycle Regulation: European regulation EN 300 220 limits 868 MHz ISM band duty cycle to 1% (36 s/hour). At 433 MHz (used in this study), the limit is more permissive but still constrains maximum transmission frequency. With SF7 ToA of 41.2 ms and 10-minute cycles, our system uses only 0.007% duty cycle -- well within regulatory limits.

Environmental Interference: Soil moisture, vegetation density, and terrain irregularities affect LoRa propagation. The log-distance path loss model used (n = 2.7) represents urban/suburban conditions; agricultural deployments with clear line-of-sight may achieve better range, while dense vegetation could reduce effective range by 30-50% [24].

5.10. Limitations and Real-World Deployment Challenges

This section consolidates the principal limitations that must be addressed before field deployment:

Biofilm Formation Delay: Soil MFCs require 2-6 weeks for stable biofilm formation on the anode electrode [1,3]. During this maturation period, power output is significantly below steady-state values, and the system would require temporary battery supplementation or extended sleep-only operation.

Soil Variability: MFC performance is highly sensitive to soil composition, organic carbon content, moisture level, and microbial community structure [2,5]. Sandy soils with low organic matter may produce 50-70% less power than the loamy agricultural soils assumed in this simulation. Site-specific calibration would be necessary.

Sensor Drift: Long-term deployment (months to years) introduces sensor drift in pH electrodes, capacitive moisture probes, and gas sensors. Without periodic recalibration, measurement accuracy degrades, potentially causing the Isolation Forest to produce false positives or miss genuine anomalies.

Energy Instability: Seasonal temperature variations (0-40 C in temperate climates) directly affect both MFC electrochemical kinetics and supercapacitor ESR. Winter operation may reduce MFC output by 40-60% compared to summer baselines [11], requiring dynamic duty-cycle extension to maintain energy balance.

Synthetic Data Limitation: The ML models are trained entirely on simulation-generated data. While Scenario A/B separation mitigates circular validation, the models may be overfit to the assumptions of the Nernst-Monod simulation. Real-world sensor noise patterns, electromagnetic interference, and non-Gaussian outliers are not captured.

5.11. Practical Deployment Scenario

To illustrate real-world applicability, we describe a reference deployment scenario for precision agriculture soil monitoring:

Target Area: A 10-hectare arable field in a temperate agricultural region. Node Placement: 16 sensor nodes arranged in a 4x4 grid with 80 m spacing, each node comprising one soil-buried MFC electrode pair, the ESP32+LoRa sensor assembly mounted at ground level, and four environmental sensors (moisture, temperature, pH, gas). Gateway: A single solar-powered LoRa gateway with WiFi/4G uplink positioned at the field edge, covering all 16 nodes within a 600 m radius (well within SF7 range of 1.2 km). Data Resolution: At 10-minute duty cycles with 82% TX success rate, each node generates approximately 118 data points/day, yielding 1,888 spatiotemporal measurements/day across the field. This density enables detection of localised anomalies (e.g., a drainage failure affecting 2-3 nodes) within 20 minutes of onset.

Estimated deployment cost: 16 nodes x $40 BOM + 1 gateway (~$80) + installation labour = approximately $800 total, with zero recurring energy or battery replacement costs. Battery-dependent commercial sensor nodes of equivalent capability typically carry additional per-node recurring costs for battery servicing, maintenance, and connectivity subscriptions [21], providing a total-cost-of-ownership advantage for the proposed battery-free architecture that grows with deployment scale and duration.

6. Conclusions

This paper presented a simulation-grounded, hardware-aware co-design framework for a microbial fuel cell energy harvesting system integrating an ESP32-based embedded sensor node and an adaptive embedded ML layer for autonomous environmental monitoring. A Simulation-Emulation-Prototype pipeline distinguishes this work from purely simulation-driven prior art: power-trace emulation confirms the simulation model is conservative (MAPE = 8.5%), providing a hardware-grounded lower-bound guarantee. Key findings are:

* The adaptive duty-cycle strategy (RF decision model approximating the optimal threshold controller) reduces average energy consumption by 47.5% vs. always-transmit (0.413 vs 0.787 mJ/cycle), enabling self-sustaining operation within the MFC power envelope.

* Isolation Forest anomaly detection achieves F1-score = 0.955 +/- 0.022 (5-fold CV, n = 50/class, ROC-AUC = 0.999) without requiring labelled training data, making it directly deployable on new installations.

* LoRa SF7 (41.2 ms ToA, 16.2 uJ per packet) is the optimal default spreading factor for sub-1.2 km links; SF9 provides the best range-energy trade-off for emergency transmissions.

* The full on-device inference pipeline (Isolation Forest ~115 KB + decision model ~45 KB, ~12 ms combined latency) runs within the 520 KB SRAM constraint of the ESP32, consistent with TinyML deployment guidelines [19,20].

* Parametric robustness testing confirms energy-positive operation in 81.5% of +/-30% MFC parameter variations; a 200-trial Monte Carlo including hardware imperfections yields 75.0% energy-positive, demonstrating graceful degradation under combined stress.

* Power-trace emulation (literature-calibrated MFC profile, V_oc_init = 0.80 V, tau = 180 s) validates the energy simulation model (MAPE = 8.5%) vs. the flat-nominal baseline, confirming that simulation constitutes a conservative lower bound.

* The open-source Simulation-Emulation-Prototype framework enables reproducible, risk-reduced hardware development with a structured path to physical validation.

Future work will prioritise: (1) benchtop validation using a laboratory soil MFC or programmable power supply emulator to compare measured vs. simulated energy budgets; (2) replacement of the rule-based Random Forest with reinforcement learning for online policy adaptation; (3) concept drift mitigation through sliding-window retraining for long-term field deployments; and (4) multi-node scalability evaluation with LoRa collision modelling and SF allocation optimisation.

This study intentionally focuses on pre-prototyping validation, providing a reproducible and extensible simulation framework for future hardware implementation. All source code is released open-source to facilitate independent replication and adaptation to alternative MFC chemistries, sensor configurations, or communication protocols.

Author Contributions:

K.K.: Conceptualisation, Methodology, Software, Validation, Writing--Original Draft. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement:

All simulation code, firmware, and dashboard source files are openly available at https://github.com/kahyakursat1-cloud/mfc-aiot-sensor (accessed 28 April 2026). Data generated during this study are contained within the paper.

Conflicts of Interest:

The authors declare no conflict of interest.

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